

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks:100

Class: S.E.

Name of the Course: Data Structures

Course Code: UCEC303

Duration: 3.00Hrs.

Semester: III

Branch: Computer

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	What are the different ways are available to represent the graph? Give their C declaration along with example.	10
Q 1 (b)	What is an AVL tree? Construct AVL tree for the following data. Mention the type of rotation for each case. 50, 25, 10, 5, 7, 3, 30, 20, 8, 15	10
Q2 (a)	What is heap? Consider the following numbers, 15, 19, 10, 7, 17, 16. Sort these numbers using Heap Sort. OR Write a C program to implement infix to prefix conversion.	10 10
Q2 (b)	What is indexed sequential search? Write a C program to implement it. OR What is Hashing? What do you mean by collision use linear probing and mod division to store the values in Hash table of SIZE 10. The numbers are 10, 15, 13, 25, 27, 37, 54, 65, 63	10 10
Q3 (a)	Explain Huffman algorithm with an example. OR With reference to Singly linked list write a C program to sort the linked list in ascending order	10 10
Q3 (b)	Explain Quick sort? For the following numbers show o/p after each pass when Quick sort is applied on it. : 10, 20, 40, 30, 90, 80, 70, 50 OR Explain Merge Sort. Apply Merge sort on the following numbers and show output after each pass : 10, 20, 40, 30, 90, 80, 70, 50	10 10
Q4 (a)	What do you mean by Recursion? State the general guideline for converting any recursive function to non-recursive function with example	10
Q4 (b)	Explain Stack as an ADT? Implement stack operations using linked list.	10
Q5 (a)	Write a program in C to check whether the given string is Balanced Parenthesized or not?	10
Q5 (b)	What is Binary threaded search tree? Write a function to insert a node in Threaded Binary search tree and performing in order traversal of it?	10